BUSINESS QUESTIONS WITH ANSWERS

1. Which brands have the highest average bike prices?

The analysis identifies the top 10 brands based on their average bike prices. Premium brands generally have higher average prices.

2. Which brands offer the highest average mileage?

The brand-wise mileage analysis identifies the top 10 brands with the highest average mileage among bikes with available mileage data.

3. Which bikes have the highest customer ratings?

The analysis displays the top 10 highest-rated bikes based on user ratings.

4. Is there a relationship between bike mileage and price?

The scatter plot helps understand the relationship between mileage and price. It shows how bike prices vary across different mileage values.

5. How does engine capacity affect bike price?

The analysis shows a strong positive relationship between engine capacity and price. Higher engine capacity bikes generally have higher prices.

6. Which are the top 10 most expensive bikes?

The analysis identifies the 10 bikes with the highest prices in the dataset.

7. Which are the top 10 most affordable bikes?

The analysis identifies the 10 bikes with the lowest prices in the dataset.

8. How are bike prices distributed in the market?

The price distribution histogram shows how bikes are spread across different price ranges.

9. Which engine capacities are most common?

The engine capacity distribution histogram shows the spread of engine capacities among bikes with available data.

10. Which brands have the highest number of bikes in the dataset?

The brand-wise bike count analysis identifies the top 10 brands based on the number of bike records collected.

BUSINESS INSIGHTS

1. **Market Coverage:** The dataset contains 528 bike records from 56 different brands.
2. **Average Bike Price:** The average bike price in the dataset is approximately ₹6,62,212.87.
3. **Average Mileage:** The average mileage among bikes with available mileage data is 38.27 km/l.
4. **Most Expensive Bike:** Ducati Panigale V4 Lamborghini is the most expensive bike in the dataset, priced at ₹1,00,00,000.
5. **Most Affordable Bike:** TVS XL 100 Heavy Duty is the most affordable bike, priced at ₹47,929.
6. **Mileage and Price Relationship:** The correlation between mileage and price is -0.474, indicating a moderate negative relationship in the available data.
7. **Engine Capacity and Price:** The correlation is 0.764, indicating a strong positive relationship. Bikes with higher engine capacity generally have higher prices.
8. **Data Availability:** Mileage information is available for 208 bikes, while engine capacity information is available for 339 bikes. Missing specifications may affect comparisons.

RECOMMENDATIONS

1. **Affordable Bike Segment:** TVS, Hero and other budget-friendly brands should focus on customers looking for low-cost bikes.
2. **Premium Bike Segment:** Ducati, BMW and Harley-Davidson can target customers interested in premium and high-performance bikes.
3. **Mileage Improvement:** Brands such as Hero, TVS, Yamaha and Bajaj can highlight fuel efficiency to attract mileage-conscious customers.
4. **Engine Capacity:** Since engine capacity has a strong positive relationship with bike price, brands can focus on offering different engine options for different customer segments.
5. **Customer Ratings:** Companies should promote highly rated bikes and also consider the number of customer ratings before making comparisons.
6. **Data Quality:** Complete information about mileage, engine capacity, kerb weight and user ratings should be collected to improve market analysis.
7. **Product Strategy:** Bike manufacturers should maintain a balanced range of Budget, Mid-Range, Premium and Luxury bikes to meet different customer needs.